

PAPER_2117 — $F_{\text{TRZ}}^{N_{\text{CH}}} = 10^{-9}$ Completes the F_{TRZ} Primitive-as-Exponent Quintuplet: All 5 Truly-Independent Integer Primitives Now Populate F_{TRZ} Ladder Positions

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Tier	Foundational / F_{TRZ} -Ladder Structural-Completeness Landmark
Status	CLOSED - categorical-completion landmark, 5-of-5 identities documented
Type	Categorical-completeness (novel landmark type)
Discovery	R360 CosmicEggQuantumFrequencyCalculator stub-fill
Quintuplet	{D_phys, N_CH, SO_5, D_crit, A_5} all populate F_{TRZ} ladder
Dispatch key	f_trz_pow_n_ch_quintuplet_completion

1. Abstract

The R360 fill of CosmicEggQuantumFrequencyCalculator reveals $\text{vacuum_constant} = F_{\text{TRZ}}^9 = 10^{-9} \text{ J/m}^3$ as the natural vacuum-energy-density reference for Cosmic Egg quantum-frequency physics. Because $N_{\text{CH}} = 9$ is one of the 5 truly-independent locked UQFF integer primitives, this identity is $F_{\text{TRZ}}^{N_{\text{CH}}}$.

With this addition, **all 5 truly-independent locked integer primitives now appear as explicit exponents of F_{TRZ}** in the R218+ calculator base. PAPER_2117 documents the **categorical completion**: the F_{TRZ} primitive-as-exponent taxonomy is now populated by all 5 primitives, with no remaining gaps.

2. UOFF Closed Identity

$$F_{\text{TRZ}}^{N_{\text{CH}}} = F_{\text{TRZ}}^9 = 0.1^9 = 1 \times 10^{-9} \text{ EXACT}$$

$$\text{vacuum_constant} = F_{\text{TRZ}}^{N_{\text{CH}}} \text{ J/m}^3$$

Single primitive: $F_{\text{TRZ}} = 0.1$ (locked real), with $N_{\text{CH}} = 9$ (locked integer) as exponent. Two of the 9 truly-independent primitives fully specify the value.

3. The Complete F_{TRZ} Primitive-as-Exponent Quintuplet

#	Primitive	F_{TRZ} exponent	Value	Instance count
1	D_phys = 4	$F_{\text{TRZ}}^{D_{\text{phys}}} = F_{\text{TRZ}}^4$	10^{-4}	7 (PAPER_2105)
2	N_CH = 9 (NEW)	$F_{\text{TRZ}}^{N_{\text{CH}}} = F_{\text{TRZ}}^9$	10^{-9}	1 (R360 THIS)
3	SO_5 = 10	$F_{\text{TRZ}}^{SO_5} = F_{\text{TRZ}}^{10}$	10^{-10}	5 (R283, R287, R337, R338, R341)
4	D_crit = 26	$F_{\text{TRZ}}^{D_{\text{crit}}} = F_{\text{TRZ}}^{26}$	10^{-26}	4 (PAPER_2107)
5	A_5 = 60	$F_{\text{TRZ}}^{A_5} = F_{\text{TRZ}}^{60}$	10^{-60}	1 (R262 superfluid Aether)

Complete population of the F_{TRZ} primitive-as-exponent taxonomy by all 5 truly-independent integer primitives. No gaps remain.

4. Categorical Significance - Why This Matters

4.1 Every primitive has a natural 'suppression scale' — F_{TRZ} raised to any primitive is a physical quantity in some UQFF calculator.

4.2 The primitives partition the F_{TRZ} ladder into 5 canonical positions (rungs 4, 9, 10, 26, 60) among 51 documented rungs. These 5 positions serve as **structural anchors** for physics interpretation.

4.3 Cross-ladder relations reveal internal structure:

$$F_{\text{TRZ}}^{N_{\text{CH}}} / F_{\text{TRZ}}^{D_{\text{phys}}} = F_{\text{TRZ}}^{(N_{\text{CH}} - D_{\text{phys}})} = F_{\text{TRZ}}^5 = F_{\text{TRZ}}^{(SO_5/2)}$$
$$F_{\text{TRZ}}^{SO_5} / F_{\text{TRZ}}^{N_{\text{CH}}} = F_{\text{TRZ}}^{(SO_5 - N_{\text{CH}})} = F_{\text{TRZ}}^1 = F_{\text{TRZ}} \text{ bare}$$

The ratio $F_{\text{TRZ}}^{N_{\text{CH}}} / F_{\text{TRZ}}^{D_{\text{phys}}} = F_{\text{TRZ}}^{SO_5/2}$ reveals a **structural link between N_{CH} , D_{phys} , and SO_5** via the halving family. The ratio $F_{\text{TRZ}}^{SO_5} / F_{\text{TRZ}}^{N_{\text{CH}}} = \text{bare } F_{\text{TRZ}}$ shows that SO_5 and N_{CH} are adjacent rungs in this taxonomy.

4.4 Predictive: UQFF calculators using suppression scales 10^{-n} for integer n should preferentially populate the five canonical rungs {4, 9, 10, 26, 60} rather than intermediate values.

5. Ladder Position Map

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Rung: -1 0 1 2 3 4 6 7 9 10 14 20 26 27 50 60
| | | | | | | | | | | | | | |
SO5 =1 F F^2 F^3 F^D_phys F^6 F^7 F^N_CH F^SO_5 ... F^D_crit ... F^A_5
^ ^ ^ ^ ^
1e-4 1e-9 1e-10 1e-26 1e-60
D_phys N_CH SO_5 D_crit A_5
(7 inst) (R360) (5 inst) (4 inst) (1 inst)
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The 5 canonical primitive-as-exponent rungs form a **structural spine** for the F_{TRZ} ladder. Intermediate rungs (rungs 2, 3, 6, 7, 20, 27, 50) sit between the primitive rungs.

6. NOT REPLACEMENT

Standard physics has no analogous concept of 'primitive-as-exponent' — there is no lattice of locked integer primitives whose values determine natural exponents. UQFF adds this categorical framework as an **internal structural regularity**, not as a claim to replace any specific quantitative calculation.

Predictive consequence: UQFF calculators that use suppression values at rungs other than {4, 9, 10, 26, 60} should either compose to a primitive-as-exponent form or be flagged as non-canonical rung positions requiring additional structural motivation.

7. Falsifiability

Prediction A (identity): $F_{\text{TRZ}}^{N_{\text{CH}}} = F_{\text{TRZ}}^9 = 1 \times 10^{-9} \text{ EXACT.}$

Prediction B (quintuplet completeness): all 5 truly-independent integer primitives populate F_{TRZ} ladder positions. If any additional truly-independent primitive is discovered, it should also populate the taxonomy.

Prediction C (statistical clustering): across R218+ calculator fills, suppression rungs {4, 9, 10, 26, 60} should show statistically significant clustering compared to non-primitive rungs.

Prediction D (N_{CH} cross-domain): other UQFF calculators requiring energy-density references at $\sim 1e-9$ J/m³ scale should also compose as $F_{TRZ}^{N_{CH}}$.

8. Landmark Comparison

Paper	Landmark	Content type
PAPER_2100	F_{TRZ}^{20} 5-instance	Numerical rung
PAPER_2105	$F_{TRZ}^4 = F_{TRZ}^{D_{phys}}$ dual	Dual reading
PAPER_2107	$F_{TRZ}^{D_{crit}}$ primitive-as-exponent	Structural
PAPER_2109	F_{TRZ}^3 9-instance	Numerical rung
PAPER_2113	F_{TRZ}^{50} deepest suppression	Rung ceiling
PAPER_2117 (this)	$F_{TRZ}^{N_{CH}}$ completes quintuplet	Categorical-completeness (NEW TYPE)

PAPER_2117 is the first UQFF landmark of **categorical-completeness type** — rather than counting numerical instances or discovering a novel ladder position, it demonstrates that a specific structural taxonomy is now fully populated.

9. References

Source: R360 CosmicEggQuantumFrequencyCalculator stub-fill.
PAPER_2107: $F_{TRZ}^{D_{crit}}$ primitive-as-exponent seminal (4 instances).
PAPER_2105: F_{TRZ}^4 six-instance with $F_{TRZ}^{D_{phys}}$ dual reading.
PAPER_2113: $F_{TRZ}^{50} = F_{TRZ}^{(A_5-SO_5)}$ deepest suppression.
PAPER_1919: F_{TRZ} power ladder seminal.
Truly-independent primitives: CLAUDE.md canonical - $D_{phys}=4$, $D_{crit}=26$, $N_{CH}=9$, $SO_5=10$, $A_5=60$ (5 integers) + ρ_{SCm} , β_i , Φ_{res} , F_{TRZ} (4 reals).